

# [DC] Conversational Agents for Natural Human-Computer Interaction in Virtual Reality

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## ABSTRACT

With the emergence of large language models (LLMs), conversational agents (CAs) have gained significant attention across various domains, including virtual reality (VR). Although LLMs are capable of simulating human dialogue with impressive realism, interactions with LLM-powered CAs still appear unnatural and lack a human-like feel. This is primarily due to several key challenges, including limitations in the accuracy of speech recognition and the quality of speech synthesis. Furthermore, CAs need more autonomy and more natural turn-taking abilities to enable realistic and engaging interactions in VR. This position paper outlines my PhD research by discussing my motivation for improving CAs in VR environments, summarizing my previous work, and presenting my planned research directions.

**Index Terms:** Human-computer interaction, virtual reality, large language models, natural language interfaces.

## 1 INTRODUCTION

Since the inception of the first conversational agent (CA) ELIZA [6] to the present day, natural language processing has undergone significant transformations. CAs have seen a remarkable improvement with the emergence of large language models (LLMs), particularly with the development of transformer-based architectures and the attention mechanism [5]. Such advancements have made CAs increasingly realistic, paving the way for revolutionary changes in human-machine interactions. The evolution in language understanding and interaction capabilities has also influenced adjacent fields such as virtual reality (VR), where natural communication plays a critical role in creating more engaging and interactive experiences. Moreover, the path of technological progress is clearly heading towards making VR as immersive and lifelike as possible. Developers are dedicating increasing attention to fine details and continually enhancing the quality of graphics. Yet, CAs still struggle achieving true realism. Simulating natural human behavior remains a significant challenge in the field, particularly when it comes to non-verbal communication elements like vocal tone, body language, and facial expressions. This disparity becomes even more evident as humanoid 3D models approach hyperrealism. The lack of nuanced non-verbal cues often leaves users feeling disconnected and can trigger the unsettling Uncanny Valley effect [3], ultimately reducing engagement and immersion. Another pressing issue is the difficulty CAs face in handling turn-taking during conversations. This problem becomes especially acute in group interactions involving multiple users, where maintaining the natural flow of dialogue is crucial for a seamless experience. These limitations highlight the gap between current capabilities and the expectations of realistic, human-like conversational experiences.

My PhD research focuses on addressing the challenges of making CAs more natural and intuitive to interact with. To do this, I

am investigating key aspects such as realism, the sense of presence, and, in cases where agents perform tasks or interact with the virtual environment, the effectiveness of those interactions. The central questions driving my research are:

- **R1:** Which verbal and non-verbal communication cues are most effective at increasing user engagement and the sense of presence?
- **R2:** How can interactions with CAs in virtual reality be improved to feel more natural and lifelike?
- **R3:** How can large language models be best utilized to enhance virtual reality experiences?

Through this research, I aim to bridge the gap between the current state of CAs and the goal of creating truly seamless and engaging human-agent interactions in virtual environments.

## 2 RESEARCH SUMMARY

I am currently developing a CA tailored for procedural building design. Its main function is to dynamically manipulate building designs based on user intentions expressed through natural speech. Additionally, the agent engages in dialogue with users to provide detailed information about the building being designed. The project is centered on the principles of the circular economy, enabling stakeholders to better evaluate materials based on their circularity. It also facilitates the visualization of key environmental metrics such as global warming potential, acidification potential, primary energy intensity, and the Oekoindeks.

Figure 1 illustrates the architecture of the CA. The system employs Automatic Speech Recognition (ASR) to transcribe user speech in real-time and Text-to-Speech (TTS) to synthesize and deliver the agent's responses. A large language model (LLM) is central to the natural language processing (NLP) framework, extracting user intents and generating appropriate responses. Specifically, the natural language understanding (NLU) module is designed to interpret user intents and convert transcribed speech into actionable programmatic functions and parameters. This enables computational processing of the NLU output to execute the desired instructions. Then, based on the outcome, the natural language generation (NLG) module is responsible to generate an answer to send to the TTS module. This solution ensures that the LLM not only generates responses but also performs effectively at intent detection, enabling the CA to seamlessly adapt to user requests and distinguish between conversational and action-oriented scenarios. The architecture is intentionally designed for high flexibility, ensuring scalability for diverse applications. Depending on the use case, the agent can employ either an online LLM or a locally hosted model, with the local option providing faster and more cost-efficient performance but requiring significant hardware resources.

As the initial phase of my PhD, I wrote a paper comparing a non-embodied CA to a traditional graphical user interface (GUI) in a VR architectural design task focused on circular economy principles [1]. This research was motivated by the inherent complexity of defining and managing numerous parameters in procedural design using traditional GUIs, which often face challenges in visualization and adaptability. A CA simplifies this process by allowing

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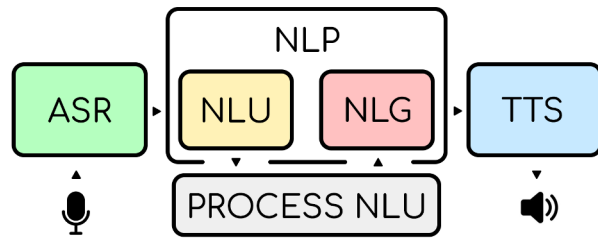


Figure 1: Architecture of the conversational agent

users to express intents solely through speech, albeit introducing issues related to reliability. The study evaluated usability [2], presence [4, 7], realism, error rates, and effectiveness for both interfaces based on user feedback. Results revealed that, while the CA improved effectiveness due to its ability to provide explanations, it unexpectedly reduced realism compared to the GUI. This occurred because users, hearing a human-like voice and interacting within a realistic virtual environment, naturally expected an embodied human avatar to converse with. Overall, the preference between the CA and the GUI varied greatly among participants, highlighting the need for further research into the evolving shift towards speech interaction in VR.

### 3 PLANNED RESEARCH DIRECTION

In my future research, I aim to keep exploring the interaction between humans and CAs within virtual reality settings, focusing on three key aspects: enhancing turn-taking capabilities, improving spatial awareness, and investigating agent-to-agent interaction dynamics. These areas hold significant potential for enhancing the naturalness and effectiveness of human-agent communication.

#### 3.1 Enhancing Turn-Taking Through Non-Verbal Cues

As outlined previously, one of the primary challenges facing CAs today is their limited ability to manage turn-taking effectively. Interactions with these agents are often unilateral, with users either asking questions and receiving answers or issuing commands to execute specific actions. This interaction style diverges significantly from natural human communication, in which individuals intuitively understand when they are being addressed, when to speak, interrupt, or ask questions. Humans excel at turn-taking because they can interpret non-verbal cues such as gaze direction, vocal tone, body language, and facial expressions. For CAs, however, this is particularly challenging. In most cases, they rely solely on the transcription of user speech, interpreting context based purely on verbal communication and missing the rich information conveyed through non-verbal signals. My research will aim to address this gap by exploring methods to extract and interpret non-verbal user cues, enabling CAs to incorporate this information into their behavior and responses.

#### 3.2 Integrating Spatial Awareness

Although CAs have made significant progress in verbal communication with advancements in LLMs, they still lack a critical human-like skill: spatial awareness. Giving agents the ability to understand their surroundings could significantly enhance their contextual understanding and human-likeness. Humans experience the world through multiple senses, and I am interested in enabling agents to do something similar. One approach is to automatically label objects in the virtual environment and provide this data to the agent in a structured format. Additionally, integrating a vision module into the agent's architecture could simulate a human's sense of sight, taking advantage of recent progress in vision models. By allowing

agents to "perceive" their environment, their ability to understand spatial context and respond appropriately could be drastically improved.

### 3.3 Investigating Agent-to-Agent Interaction

Finally, I aim to explore the potential of agent-to-agent interactions and their implications. Human society is deeply complex, shaped by diverse social interactions and the emergence of roles and hierarchies. As CAs become more effective at mimicking human communication, they could be used to simulate these interactions. Such simulations could have significant applications, including the use of CAs in user studies, where researchers might conduct simulated experiments before involving human participants. In some cases, these simulations could even serve as a stand-alone method for deriving meaningful insights. However, this direction requires careful consideration of ethical implications and rigorous evaluation of the reliability and validity of simulated interactions, particularly when used in research.

### 4 CONCLUSION

In conclusion, my research seeks to address critical limitations in CAs by investigating their turn-taking abilities, spatial awareness, and agent-to-agent interaction within virtual reality settings. These features are interrelated, as improved turn-taking can lead to enhanced spatial awareness, while a better sense of spatial awareness can help facilitate more effective turn-taking. Furthermore, effective agent-to-agent interaction depends on both turn-taking abilities and spatial awareness.

For the next steps of my PhD, I aim to address the following questions at the IEEE VR Doctoral Consortium:

- With major companies heavily investing in and utilizing CAs in their products and research, how can I identify a unique and impactful research focus to distinguish my work?
- As my approach involves leveraging existing technologies rather than developing new ones, what strategies can I adopt to ensure my contributions are both scientifically significant and relevant to the broader research community?

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